

Dynamic Pricing Competition: Pricing strategies

*Thomas Winter, Alwin Haensel
BHT Berlin*

*University of Applied Sciences
and Technology*

Thomas.Winter@bht-berlin.de



Dynamic pricing competition: Pricing strategies

Overview:

- Price matching
- Strategic Pricing Response (game-theoretic approach)
- Demand prediction and price optimization
- Monitoring of inventory

... or combinations of all strategies

Dynamic pricing competition: Pricing matching

Price Matching

- Strategy
 - Match competitor's price level
 - Simple undercutting, stay a little bit below the competitor's price
 - Possible delay factor on reacting on price changes
- Advantage
 - easy to implement
 - get's you your market share
- Disadvantage
 - risk of a price decline spiral effect
 - Limits your revenue because of focus on lowest price level

Dynamic pricing competition: Strategic Pricing Response

Strategic Price Response (Game-theoretic approach)

- Strategy
 - Try to find a Nash equilibrium by checking both your and your competitor's options
- Advantage
 - tries to avoid price wars and to maximize long-term profit.
 - considers both your and your competitor's reactions
- Disadvantage
 - complex evaluation
 - requires detailed information about the market and competitor's behavior

Dynamic pricing competition:

Demand prediction and price optimization

Demand prediction and price optimization

- Strategy
 - Try to estimate the demand model based on your price and sales observations, e.g. by regression
 - Try to optimize your price levels using the demand model
- Advantage
 - Tries to set optimal price levels to optimize the revenue
 - considers both your and your competitor's reactions
- Disadvantage
 - complex evaluation
 - requires an intensive data evaluation

Dynamic pricing competition: Monitoring of inventory

Monitoring of inventory

- Strategy
 - Monitor inventory levels and adjust prices based on sales report
- Advantage
 - Focus on your inventory in order to sell at a reasonable price level
 - Prevent early of-off-stock- situations
- Disadvantage
 - Ignores competitor actions
 - Can lead to low prices in the end when the inventory level is still high